

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

Electronic Mail (E-mail)

Definition

Electronic Mail (E-mail) is an application-layer service that enables users to exchange digital messages over the Internet.

- One of the oldest and most widely used Internet applications.
- Allows asynchronous communication.
- Messages are delivered through intermediate mail servers.
- Supports text, images, audio, video, and attachments using MIME.

How E-mail Differs from HTTP and FTP

Feature	HTTP/FTP	E-mail
Communication	Request-Response	One-way Transaction
Server	Always Running	Intermediate Mail Server
User Availability	Immediate	User can be Offline
Communication Type	Interactive	Store-and-Forward

Key Point

E-mail does not require the receiver to be online when the message is sent.

Characteristics of E-mail

- One-way communication
- Receiver may reply later
- Receiver can remain offline
- Messages are stored temporarily on mail servers
- Uses client-server architecture through intermediate servers
- Reliable message delivery

Architecture of Electronic Mail

Main Components

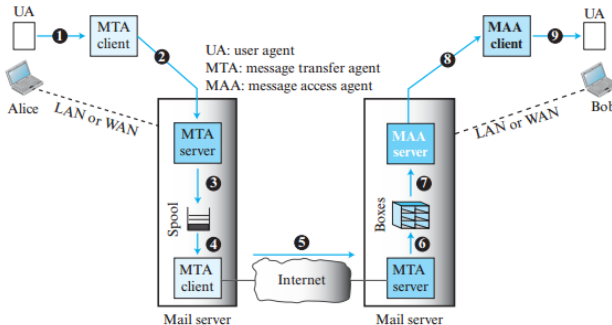
- Sender (Alice)
- Sender Mail Server
- Internet
- Receiver Mail Server
- Receiver (Bob)

Observation

Mail travels through mail servers instead of directly between users.

Electronic Mail Architecture

Figure 2.19 Common scenario



Exam Point

The architecture shows Alice, Bob, Mail Servers, User Agents (UA), Message Transfer Agents (MTA), and Message Access Agents (MAA).

Mailbox and Queue (Spool)

Each Mail Server maintains two storage areas.

Mailbox

Stores received messages for each user.

Queue (Spool)

Temporarily stores outgoing messages waiting for transmission.

- Mailbox belongs to individual users.
- Queue stores messages until delivery.

Nine-Step Mail Transfer

Mail delivery involves multiple stages.

- 1 Alice composes mail.
- 2 UA sends mail to sender mail server.
- 3 Mail stored in sender queue.
- 4 Sender MTA transfers mail.
- 5 Internet transports message.
- 6 Receiver MTA accepts mail.
- 7 Mail stored in receiver mailbox.
- 8 Bob requests messages.
- 9 Bob reads mail.

Three Important Agents

Agent	Function
User Agent (UA)	Compose, send, receive, reply and forward messages
Message Transfer Agent (MTA)	Transfers mail between mail servers using SMTP
Message Access Agent (MAA)	Retrieves mail from mailbox using POP3 or IMAP4

Remember

UA → User Interface

MTA → Mail Transfer

MAA → Mail Retrieval

Why Can't Bob Receive Mail Directly?

Problem

The receiver cannot directly receive e-mails from the sender.

Reasons

- Bob may switch off his computer.
- Bob may disconnect from the Internet.
- Running a mail server continuously on every user's computer is impractical.
- Mail servers store messages until the receiver retrieves them.

Exam Point

Mail servers allow asynchronous communication.

Push and Pull Communication

Push Protocol	Client sends data to the server.
Example	SMTP
Pull Protocol	Client retrieves data from the server.
Example	POP3 and IMAP4

Key Point

SMTP pushes mail to servers.

POP3/IMAP4 pull mail from servers.

User Agent (UA)

Definition

A User Agent (UA) is software that allows users to compose, send, receive, reply to, and forward e-mails.

Functions

- Compose messages
- Read received mail
- Reply and Forward
- Organize local mailbox
- Delete unwanted messages



Types of User Agents

Two categories of User Agents are available.

- 1 Command-driven User Agent
- 2 GUI-based User Agent

Examples

mail, pine, elm, Outlook, Eudora

Command-driven User Agents

Characteristics

- Keyboard-based interface
- Single-character commands
- Mostly used in UNIX systems
- Lightweight and efficient

Examples

- mail
- pine
- elm

GUI-based User Agents

Characteristics

- Graphical User Interface
- Icons and Menus
- Easy to compose and organize mail
- Mouse and Keyboard interaction

Examples

- Microsoft Outlook
- Eudora

Sending an E-mail

A User Agent creates two components.

Envelope

Contains sender address, receiver address and routing information.

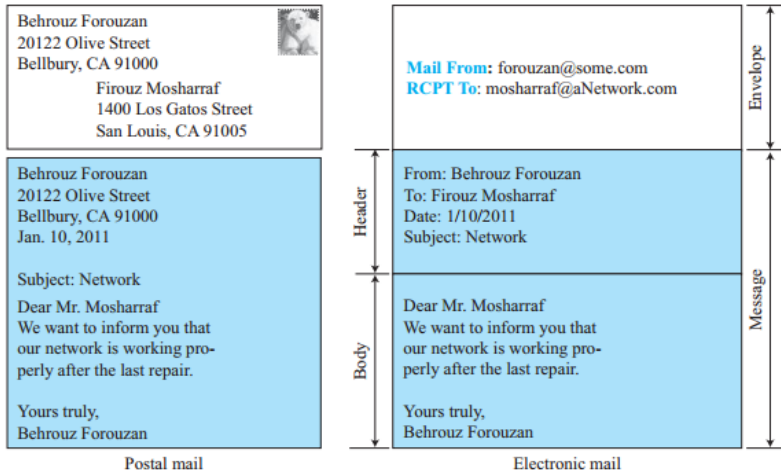
Message

Contains Header and Body.

Message Structure

Component	Purpose
Envelope	Sender and Receiver addresses
Header	From, To, Subject and other information
Body	Actual message content

Figure 2.20 *Format of an e-mail*



Receiving an E-mail

Receiving process

- 1 User Agent checks mailbox.
- 2 User receives notification.
- 3 Mail summary is displayed.
- 4 User selects required message.
- 5 Complete message is displayed.

Information shown

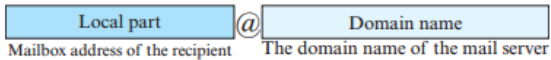
- Sender
- Subject
- Date

General Format

`local-part@domain-name`

- Local Part identifies the user's mailbox.
- Domain Name identifies the mail server.
- Every e-mail address must be unique.

Figure 2.21 *E-mail address*



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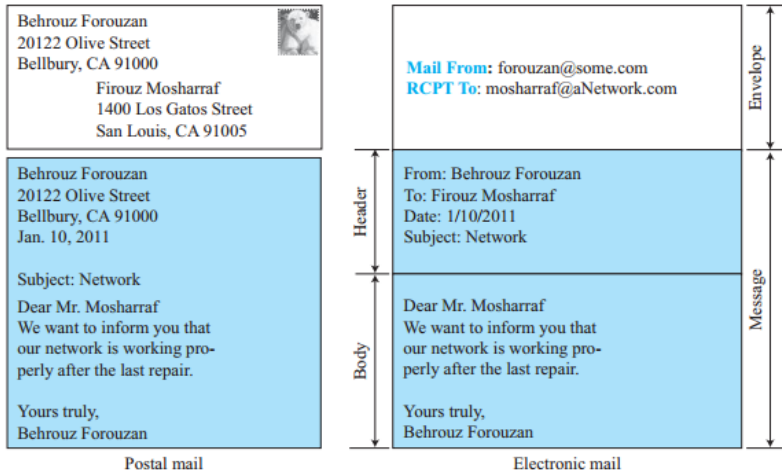
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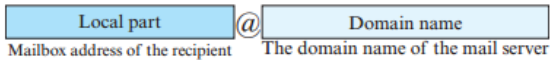
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Connection Termination

After successful mail transfer, the SMTP session is terminated.

- 1 Client sends the **QUIT** command.
- 2 Server responds with **221 Service Closing**.
- 3 TCP connection is terminated.

Key Point

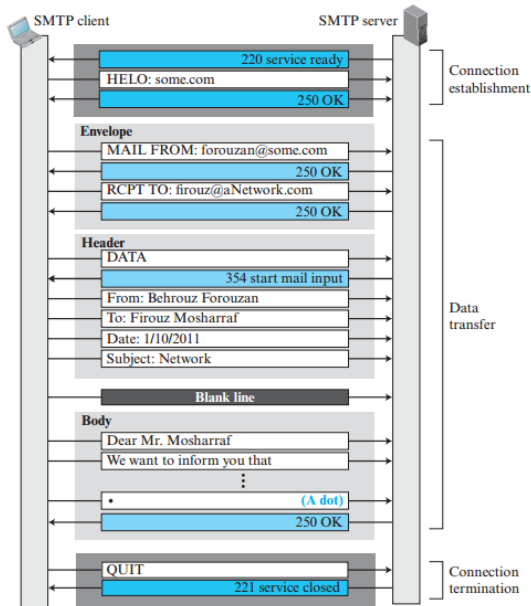
Every SMTP session ends with the **QUIT** command.

SMTP Mail Transfer Process

The complete SMTP process consists of:

- Connection Establishment
- Sender Identification
- Recipient Identification
- Message Transfer
- Connection Termination

Figure 2.23 Example 2.13



Message Access Agent (MAA)

Definition

A Message Access Agent (MAA) allows users to retrieve e-mails stored on the mail server.

Popular protocols

- POP3 (Post Office Protocol Version 3)
- IMAP4 (Internet Message Access Protocol Version 4)

Key Point

SMTP pushes mail to the server.

MAA protocols pull mail from the server.

POP3 (Post Office Protocol Version 3)

Features

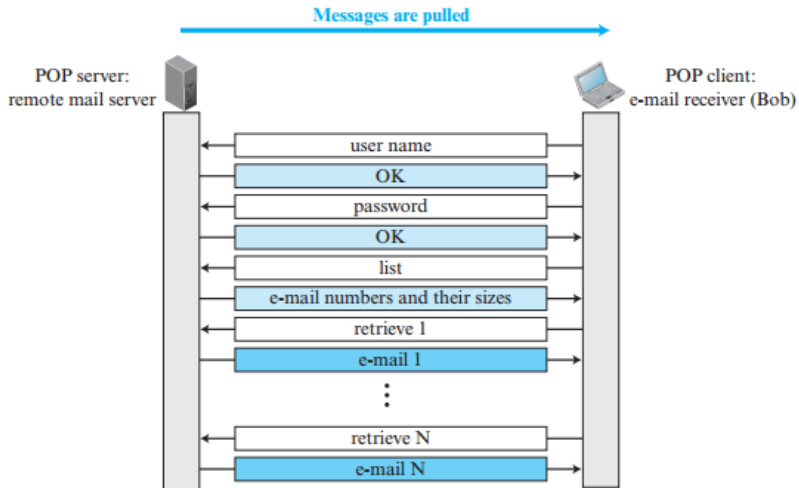
- Simple mail access protocol
- Uses TCP Port 110
- Downloads messages from server
- Client authenticates using username and password
- Suitable for single-device access

POP3 performs the following operations:

- ① Client connects to mail server.
- ② User authentication.
- ③ Mailbox is accessed.
- ④ Messages are listed.
- ⑤ Selected messages are downloaded.



Figure 2.24 POP3



POP3 Modes

Delete Mode	Messages are deleted from the server after download.
Keep Mode	Messages remain on the server after download.

Applications

- Delete Mode - Personal desktop computer
- Keep Mode - Laptop, mobile devices, multiple devices

IMAP4 (Internet Mail Access Protocol Version 4)

Definition

IMAP4 is a more advanced message access protocol than POP3.

Advantages

- Mail remains on server.
- Supports folders.
- Supports synchronization.
- Suitable for multiple devices.
- Allows partial download.

POP3 vs IMAP4

Feature	POP3	IMAP4
Mail Storage	Local PC	Server
Folders	No	Yes
Multiple Devices	Limited	Excellent
Partial Download	No	Yes
Synchronization	No	Yes

MIME (Multipurpose Internet Mail Extensions)

Need for MIME

SMTP can transfer only 7-bit ASCII text.

MIME extends e-mail by supporting:

- Images
- Audio
- Video
- PDF Documents
- Binary Files
- Multiple Languages



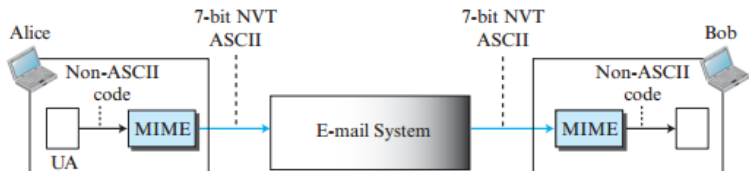
How MIME Works

MIME converts non-ASCII data before transmission.

- 1 Original multimedia file
- 2 MIME Encoding
- 3 SMTP Transmission
- 4 MIME Decoding
- 5 Original file restored



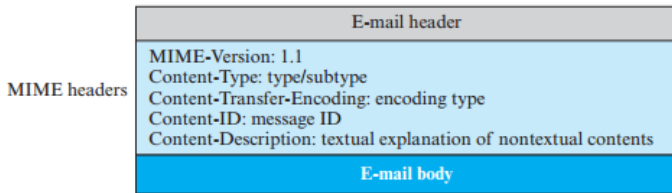
Figure 2.25 *MIME*



MIME adds additional headers to the standard e-mail header.

- MIME-Version
- Content-Type
- Content-Transfer-Encoding
- Content-ID
- Content-Description

Figure 2.26 *MIME header*



MIME Content Types

Type	Example Subtype
Text	Plain, HTML
Multipart	Mixed, Alternative
Message	RFC822
Image	JPEG, GIF
Video	MPEG
Audio	Basic
Application	PostScript, Octet-stream

Purpose

Specifies the type of data carried in the e-mail body.

Content Transfer Encoding

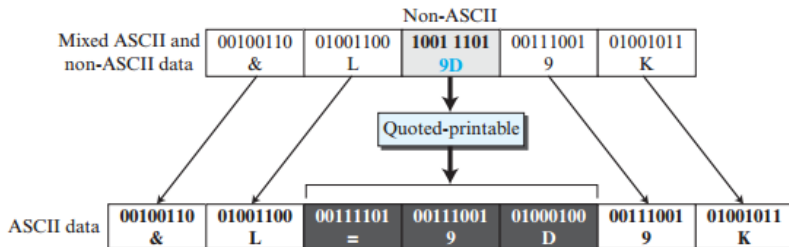
Common encoding methods supported by MIME.

Encoding	Purpose
7-bit	Standard ASCII
8-bit	Extended ASCII
Binary	Binary Data
Base64	Binary to ASCII conversion
Quoted-Printable	Mostly ASCII with few special characters

Working

- Binary data is divided into 6-bit groups.
- Each 6-bit group is mapped to an ASCII character.
- Suitable for sending images and documents.

Figure 2.28 *Quoted-printable*

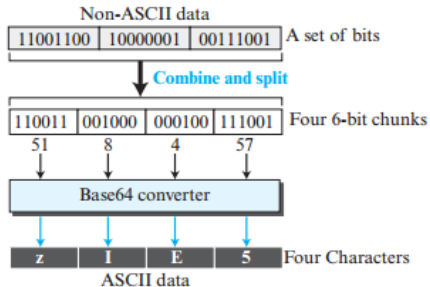


Quoted-Printable Encoding

Used when the message contains mostly ASCII characters.

- ASCII characters are transmitted unchanged.
- Non-ASCII characters are represented using '=' followed by hexadecimal digits.
- Reduces encoding overhead.

Figure 2.27 *Base64 conversion*



Web-Based E-mail

Examples

- Gmail
- Yahoo Mail
- Outlook.com

Characteristics

- Browser-based access
- No dedicated mail client required
- Uses HTTP/HTTPS for user interaction



Working of Web-Based Mail

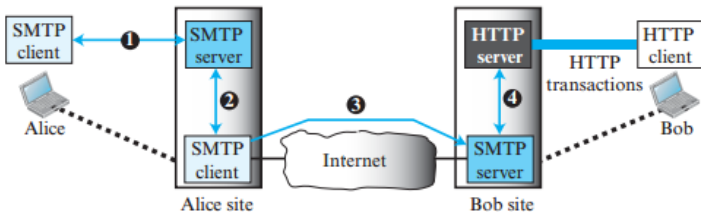
Case I

Traditional sender communicates with a web-mail user.

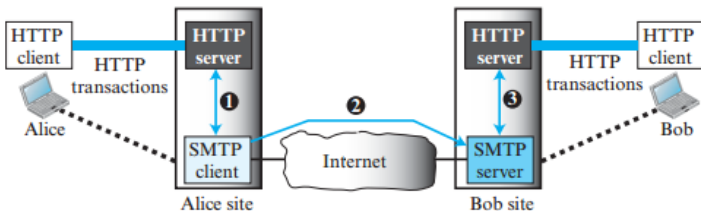
Case II

Both sender and receiver use web-mail services.

Figure 2.29 *Web-based e-mail, cases I and II*



Case 1: Only receiver uses HTTP



Case 2: Both sender and receiver use HTTP

E-mail Security

The basic SMTP protocol does not provide security.

Security is provided using:

- Pretty Good Privacy (PGP)
- Secure/Multipurpose Internet Mail Extensions (S/MIME)

Security services include:

- Confidentiality
- Authentication
- Integrity
- Digital Signatures

Summary

- Electronic Mail is an asynchronous communication system.
- User Agent (UA) provides user interface.
- SMTP transfers e-mails between servers.
- POP3 and IMAP4 retrieve e-mails.
- MIME enables multimedia transmission.
- Web-based mail uses HTTP/HTTPS.
- PGP and S/MIME improve e-mail security.

Topics Covered

- Electronic Mail Architecture
- User Agent (UA)
- Message Transfer Agent (SMTP)
- Message Access Agent (POP3, IMAP4)
- MIME
- Web-Based Mail
- E-mail Security

Thank You